

Moving around

Hold spacebar then click and drag to move around.

Find it



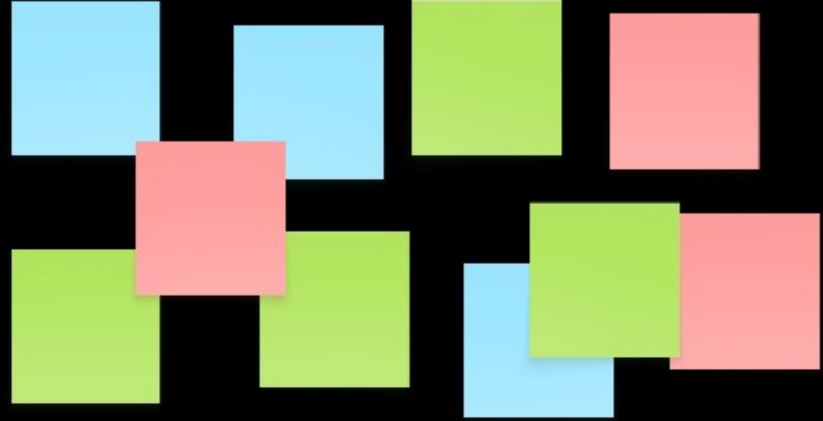
Hold **spacebar** then click and drag to move around

See it



Try it

- ① Hold **spacebar** until you see a hand
- ② Place your **cursor** over the stickies, then **click and drag**



 Edit the board when your cursor looks like a **pointer** (v). Move around with the **hand** (h).

Zooming in and out

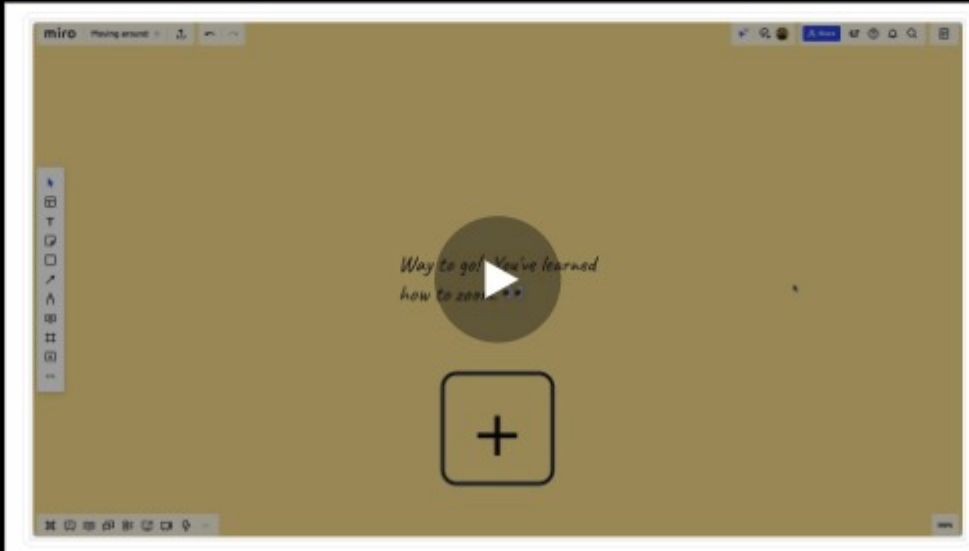
Use the plus and minus keys on your keyboard to zoom in and out.

Find it



Use the **plus** and **minus** keys on your keyboard

See it



Try it

- ① Center your screen on the small text below
- ② Use the **plus** and **minus** keys to zoom in and read the message

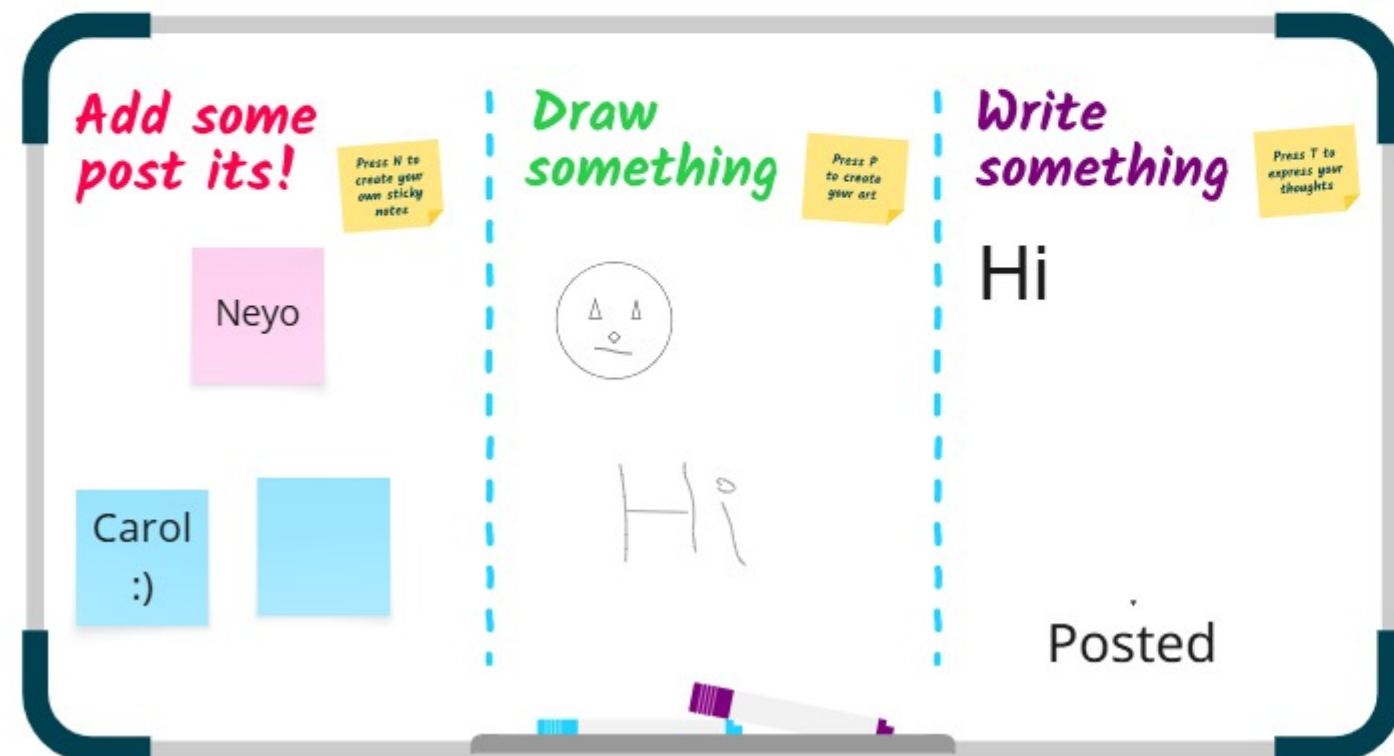


 Alternatively, use the **scroll wheel** on a mouse or **pinch** on a trackpad.

Have a play

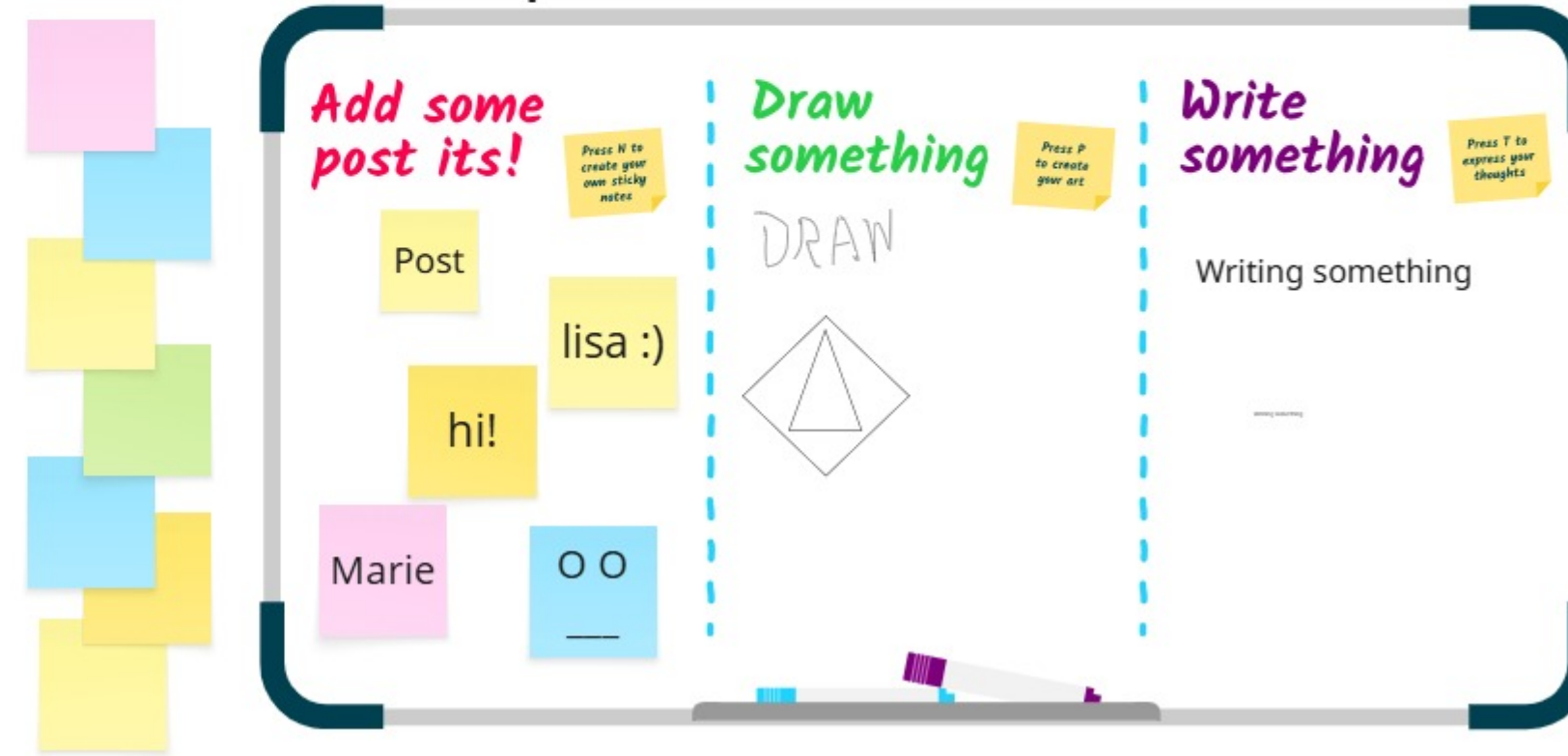
Use the tools to get familiar with adding content.

Group 1



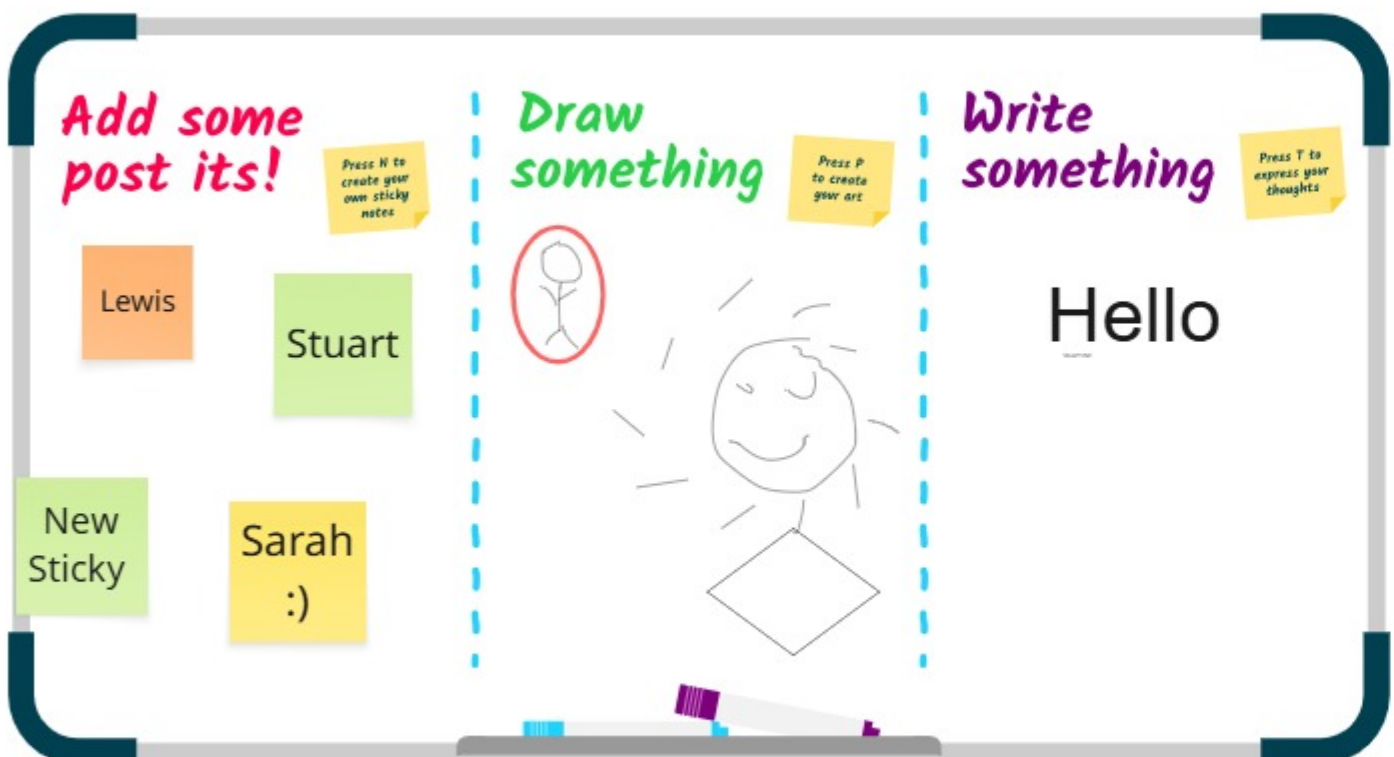
Group 1 digital whiteboard with three columns: 'Add some post its!', 'Draw something', and 'Write something'. The 'Add some post its!' column contains a pink sticky note with 'Neyo', a blue sticky note with 'Carol :)', and an empty blue sticky note. The 'Draw something' column contains a simple face drawing and the word 'Hi' written in blue. The 'Write something' column contains the word 'Hi' and the word 'Posted'.

Group 2



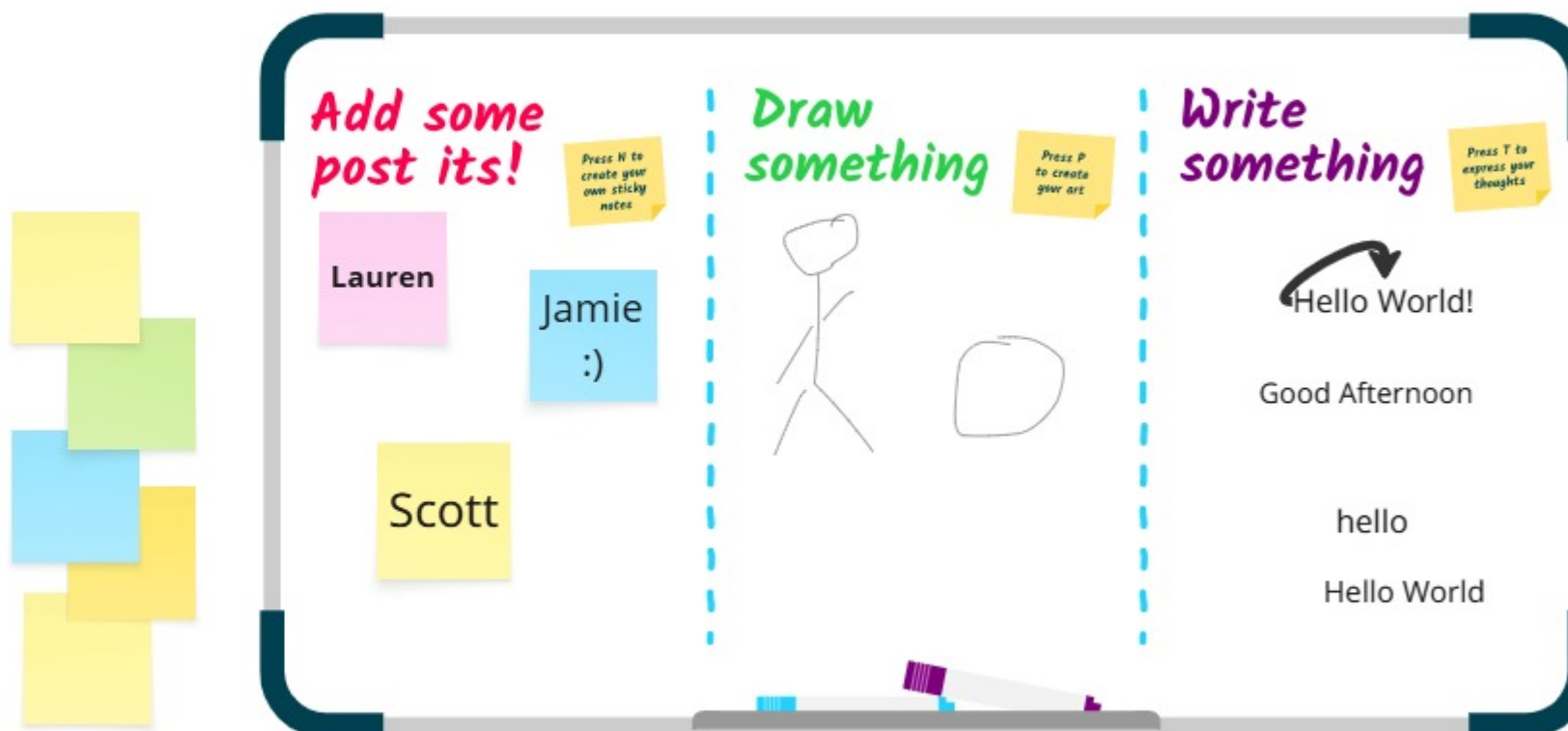
Group 2 digital whiteboard with three columns: 'Add some post its!', 'Draw something', and 'Write something'. The 'Add some post its!' column contains a pink sticky note with 'Post', a yellow sticky note with 'hi!', a pink sticky note with 'Marie', and a blue sticky note with 'lisa :)'. The 'Draw something' column contains a diamond shape with a triangle inside and the word 'DRAW' written above it. The 'Write something' column contains the text 'Writing something'.

Group 3



Group 3 digital whiteboard with three columns: 'Add some post its!', 'Draw something', and 'Write something'. The 'Add some post its!' column contains an orange sticky note with 'Lewis', a green sticky note with 'Stuart', a green sticky note with 'New Sticky', and a yellow sticky note with 'Sarah :)'. The 'Draw something' column contains a stick figure drawing and a diamond shape. The 'Write something' column contains the word 'Hello'.

Group 4



Group 4 digital whiteboard with three columns: 'Add some post its!', 'Draw something', and 'Write something'. The 'Add some post its!' column contains a pink sticky note with 'Lauren', a blue sticky note with 'Jamie :)', and a yellow sticky note with 'Scott'. The 'Draw something' column contains a stick figure drawing and a diamond shape. The 'Write something' column contains the text 'Hello World!', 'Good Afternoon', 'hello', and 'Hello World'.

Housekeeping

Rules regarding webcams

Keep your Webcam on throughout the delivery

Rules regarding mute

Keep your Microphone off throughout for privacy unless you are asked or need to respond.

Frequency of breaks

Short Breaks are scheduled every hour to support you with your learning.

Rules regarding group participation

Everyone should equally participate in Group discussion and take lead in contributing your view.

Rules regarding questions

The Golden Rule:

'There is no such thing as a stupid question'

'Even when asked by an instructor'

Please have a go at answering questions

Also:

'A question never resides in a single mind'
By asking questions, you are helping everyone out!

Anything else



Exercise

Step 1: Initial Setup

Task 1: Identify the initial state of the system. The initial state is a simple environment with a robot and a goal.

Task 2: Identify the initial state of the system. The initial state is a simple environment with a robot and a goal.

Task 3: Identify the initial state of the system. The initial state is a simple environment with a robot and a goal.

Step 2: Action Space

Task 1: Identify the action space. The action space is defined by the robot's movement capabilities.

Task 2: Identify the action space. The action space is defined by the robot's movement capabilities.

Task 3: Identify the action space. The action space is defined by the robot's movement capabilities.

Step 3: State Space

Task 1: Identify the state space. The state space is defined by the robot's position and orientation.

Task 2: Identify the state space. The state space is defined by the robot's position and orientation.

Task 3: Identify the state space. The state space is defined by the robot's position and orientation.

Step 4: Transition Function

Task 1: Identify the transition function. The transition function maps the current state and action to the next state.

Task 2: Identify the transition function. The transition function maps the current state and action to the next state.

Task 3: Identify the transition function. The transition function maps the current state and action to the next state.

Step 5: Reward Function

Task 1: Identify the reward function. The reward function maps the current state and action to a scalar reward.

Task 2: Identify the reward function. The reward function maps the current state and action to a scalar reward.

Task 3: Identify the reward function. The reward function maps the current state and action to a scalar reward.

Step 6: Policy Function

Task 1: Identify the policy function. The policy function maps the current state to a probability distribution over actions.

Task 2: Identify the policy function. The policy function maps the current state to a probability distribution over actions.

Task 3: Identify the policy function. The policy function maps the current state to a probability distribution over actions.

Step 7: Value Function

Task 1: Identify the value function. The value function maps the current state to a scalar value representing the expected return.

Task 2: Identify the value function. The value function maps the current state to a scalar value representing the expected return.

Task 3: Identify the value function. The value function maps the current state to a scalar value representing the expected return.

Step 8: Final Results

Task 1: Identify the final results. The final results are the optimal policy and value function.

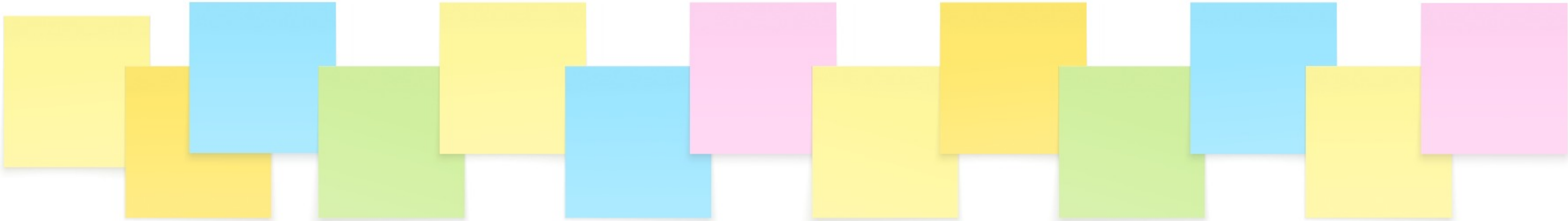
Task 2: Identify the final results. The final results are the optimal policy and value function.

Task 3: Identify the final results. The final results are the optimal policy and value function.

Entity 2

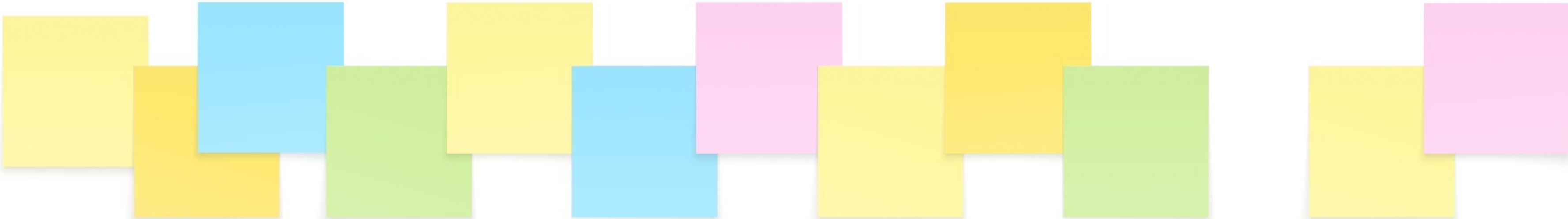
Entity 1

Entity description	Movie List		
Database	Movie		
Entity name	Movie		
Key	Attribute	Data Type	Constraints/comments
Promary Key	Movie_ID	Int	Not null
	Title	Varchar(MAX)	Not null
	Budget	money	Null allowed
	Release_Date	Date	Null Allowed, must be valid date
	Movie_Status	Varchar(20)	Not null
	Popularity	Float	Null Allowed
	Overview	Varchar (Max)	Null Allowed
	Runtime	Int	Null Allowed
	Revenue	Money	Null Allowed
	Homepage	Varchar(255)	Null Allowed
	Tagline	Varchar(Max)	Null Allowed
	Vote_Average	Decimal(5, 2)	Null Allowed
	Vote_Count	Int	Null Allowed



Entity 1

Entity description	Actor List		
Database	Movie		
Entity name	Actor		
Key	Attribute	Data Type	Constraints/comments
Primary Key	Actor_ID	Int	Not null
	Actor	Varchar(100)	Not null
	Gender	Varchar(10)	Null allowed



Teams

	Carol	Chris	Neyo									
	Lisa	Louis	Marie									
	Connor	Lewis	Sarah	Stuart								
	Jamie	Lauren	Scott									

Go to your team board


Group 1


Group 2


Group 3

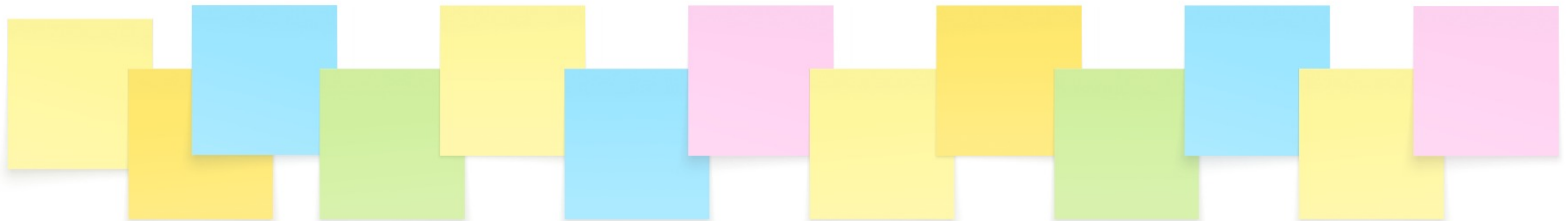

Group 4

Task 1

Identify and name potential entities from the information supplied.

There are at least 5 entities needed and 2 of these are bridging tables.

1. Director
2. Movie
3. Actor
4. MovieDirector (bridge)
5. MovieCast (bridge)

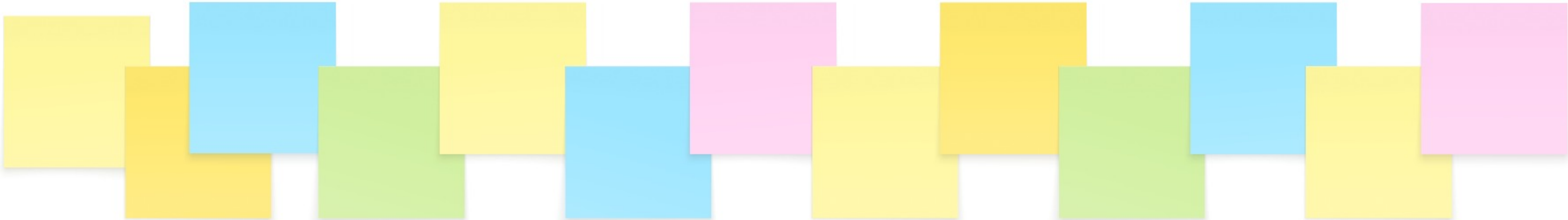


Movies and Actors Database

Director	Actor	gender	character_name	cast_order	movie_id	title	budget	homepage	overview	popularity	release_date	revenue	runtime	movie_status	tagline	vote_average	vote_count
Quentin Tarantino	Tim Roth	Male	Ted the Bellhop	43	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	65
Robert Rodriguez	Tim Roth	Male	Ted the Bellhop	44	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	65
Allison Anders	Tim Roth	Male	Ted the Bellhop	45	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	65
Alexandre Rockwell	Tim Roth	Male	Ted the Bellhop	46	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	65
Quentin Tarantino	Antonio Banderas	Male	Man	48	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	65
Robert Rodriguez	Antonio Banderas	Male	Man	49	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	65
Allison Anders	Antonio Banderas	Male	Man	50	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	65
Alexandre Rockwell	Antonio Banderas	Male	Man	51	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	65
Quentin Tarantino	Jennifer Beals	Female	Angela	52	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	65
Robert Rodriguez	Jennifer Beals	Female	Angela	53	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	65
Allison Anders	Jennifer Beals	Female	Angela	54	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hands	8	65

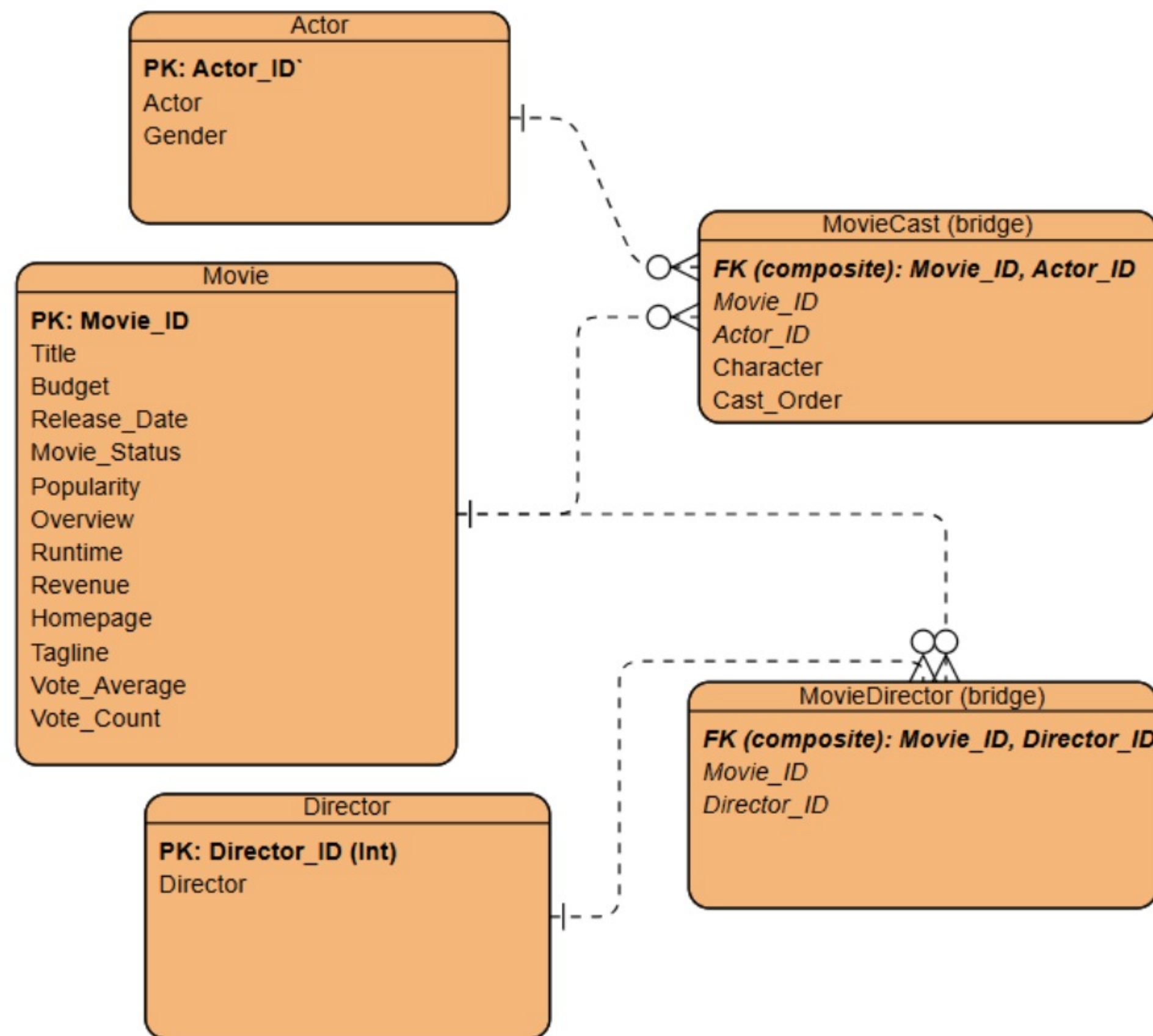
The attributes in the Movies and Actors table are:

- Director
- Actor
- gender
- character_name
- cast_order
- movie_id
- title
- budget
- homepage
- overview
- popularity
- release_date
- revenue
- runtime
- movie_status
- tagline
- vote_average
- vote_count



Task 3

Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities (tables) you identified in Task 1.

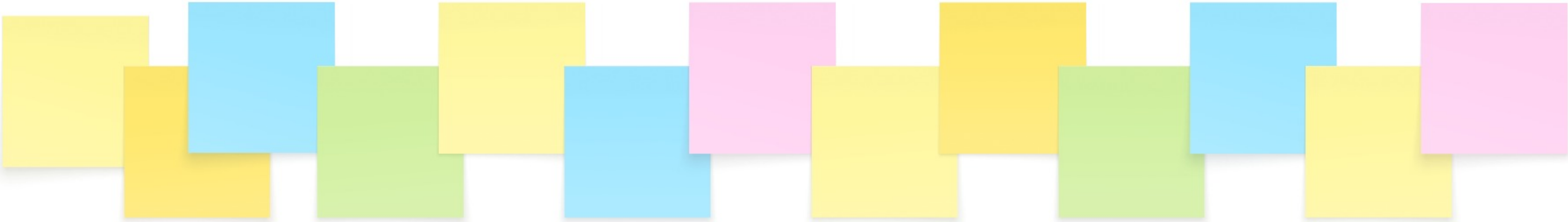


Task 2

Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity 1

Entity description	Director List		
Database	Movie		
Entity name	Director		
Key	Attribute	Data Type	Constraints/comments
Primary Key	Director_ID	Int	Not null
	Director	Vchar (50)	Not null





Exercise



Entity 2

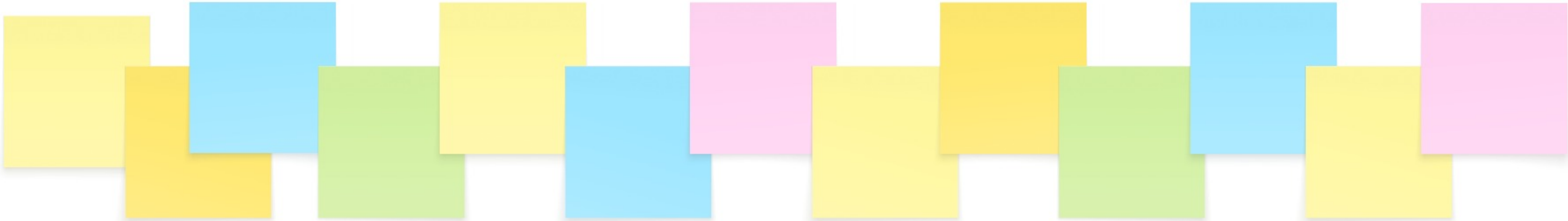
Entity 1

Entity description	Movie		
Database	Movies		
Entity name	Movie		
Key	Attribute	Data Type	Constraints/comments
Primary Key	Movie ID	Int	Not Null
	Release Date	DATE	Not Null
	Tile	VARCHAR(50)	Not Null
	Runtime	TIME	Not Null
	budget	FLOAT	Money
	tagline	VARCHAR (100)	Not Null
	revenue	FLOAT	Money
	CastOrder	INT	Not Null
	Director	VARCHAR (50)	Not Null
	Overview	VARCHAR (250)	Not Null
	Popularity	INT	Null allowed
	Homepage	VARCHAR (150)	Null allowed
	Vote Count	INT	Null allowed
	Vote Average	INT	Null allowed

Entity 3

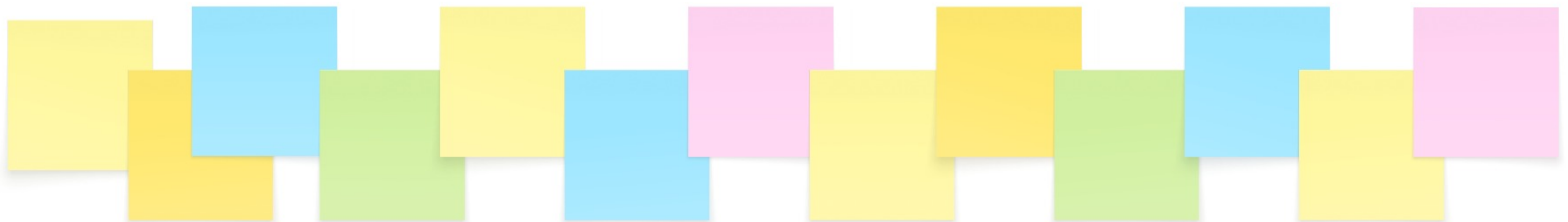
Entity 1

Entity description	Actors		
Database	Movies		
Entity name	Actor		
Key	Attribute	Data Type	Constraints/comments
Primary Key	Actor ID	Int	Not Null, Unique
	Gender	CHAR	Not Null
	Actor Name	VARCHAR(255)	Not Null



There are at least 5 entities needed and 2 of these are bridging tables.

1. Director
2. Movie
3. Actor
4. Character
5. Movie Cast - Bridge

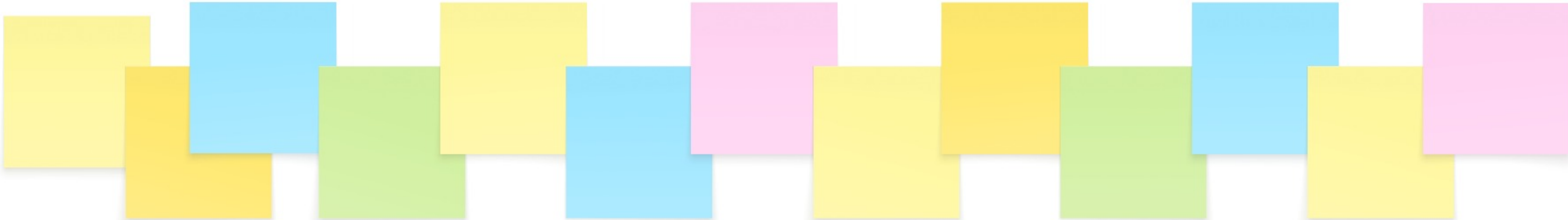


Movies and Actors Database

Director	Actor	gender	character_name	cast_order	movie_id	title	budget	homepage	overview	popularity	release_date	revenue	runtime	movie_status	tagline	vote_average	vote_count
Quentin Tarantino	Tim Roth	Male	Ted the Bellhop	43	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Tim Roth	Male	Ted the Bellhop	44	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Tim Roth	Male	Ted the Bellhop	45	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Alexandre Rockwell	Tim Roth	Male	Ted the Bellhop	46	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Quentin Tarantino	Antonio Banderas	Male	Man	48	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Antonio Banderas	Male	Man	49	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Antonio Banderas	Male	Man	50	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Alexandre Rockwell	Antonio Banderas	Male	Man	51	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Quentin Tarantino	Jennifer Beals	Female	Angela	52	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Jennifer Beals	Female	Angela	53	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Jennifer Beals	Female	Angela	54	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65

The attributes in the Movies and Actors table are:

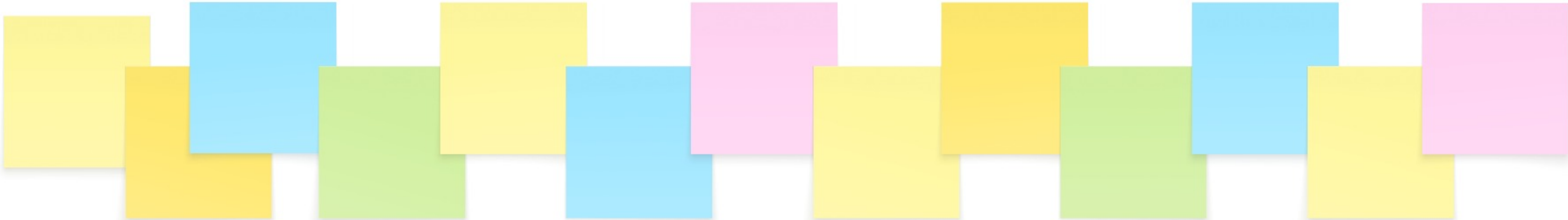
- Director
- Actor
- gender
- character_name
- cast_order
- movie_id
- title
- budget
- homepage
- overview
- popularity
- release_date
- revenue
- runtime
- movie_status
- tagline
- vote_average
- vote_count



Entity 5

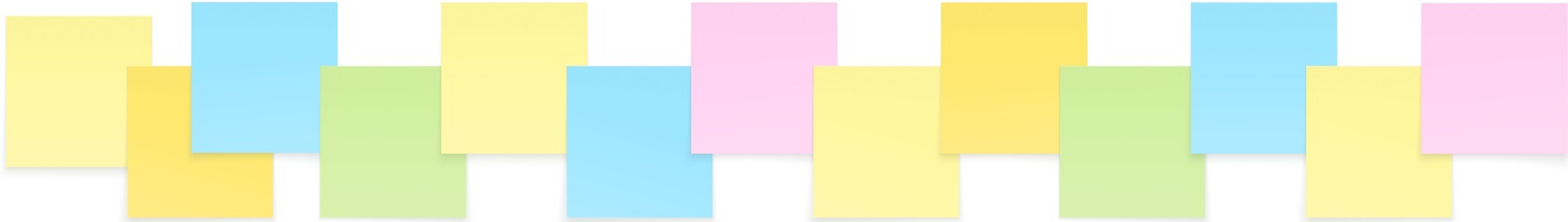
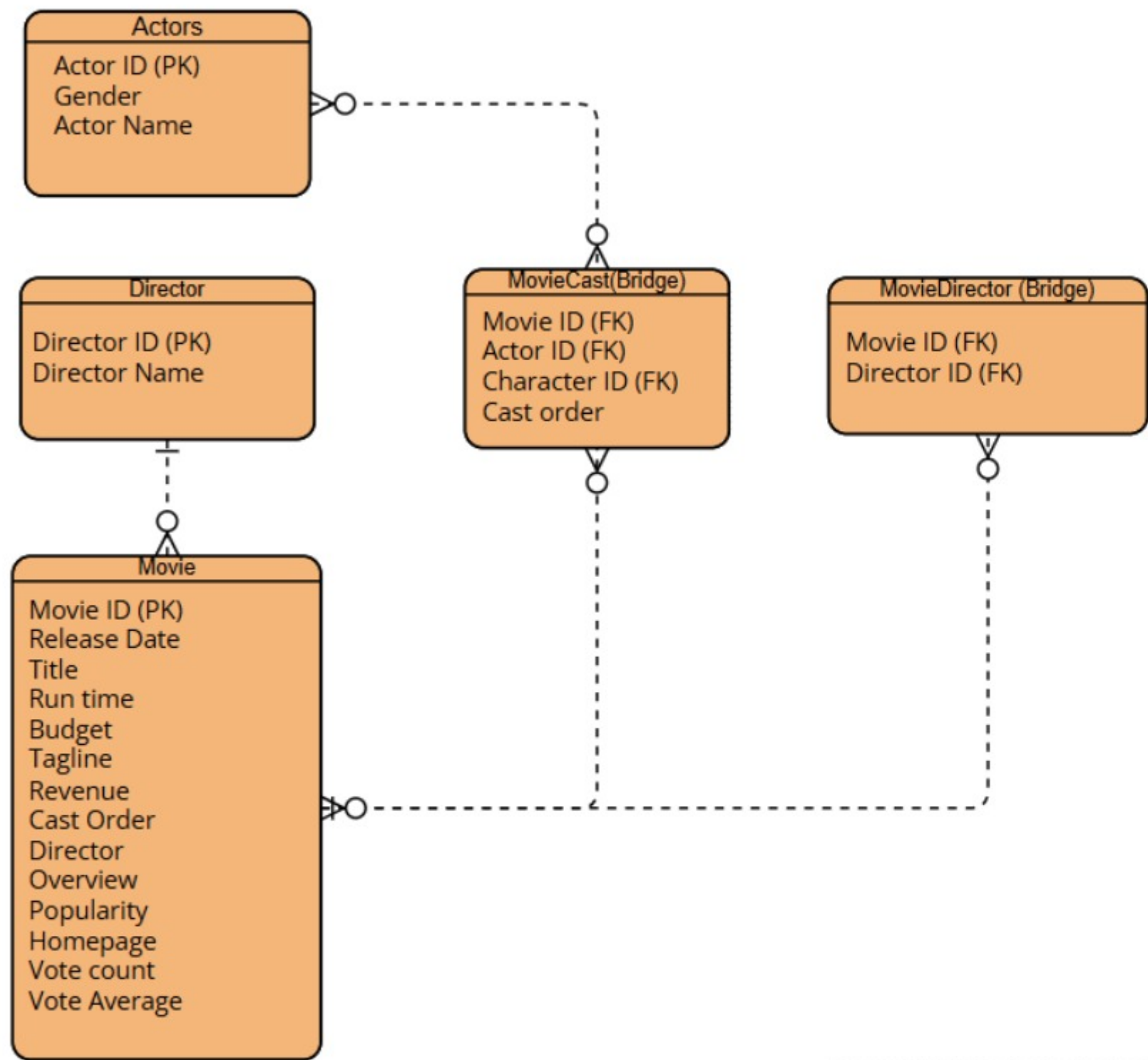
Entity 1

Entity description	Movie Casting		
Database	Movie		
Entity name	Movie Cast		
Key	Attribute	Data Type	Constraints/comments
FK	Movie ID	INT	Unique, Not Null
FK	Actor ID	INT	Unique, Not Null
FK	Character ID	VARCHAR	Unique, Not Null
	Cast Order	INT	



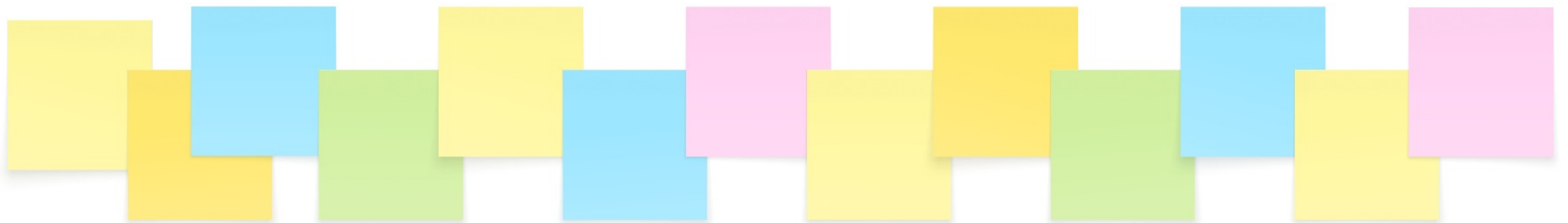
Task 3

Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities(tables) you identified in Task 1.



Entity 1

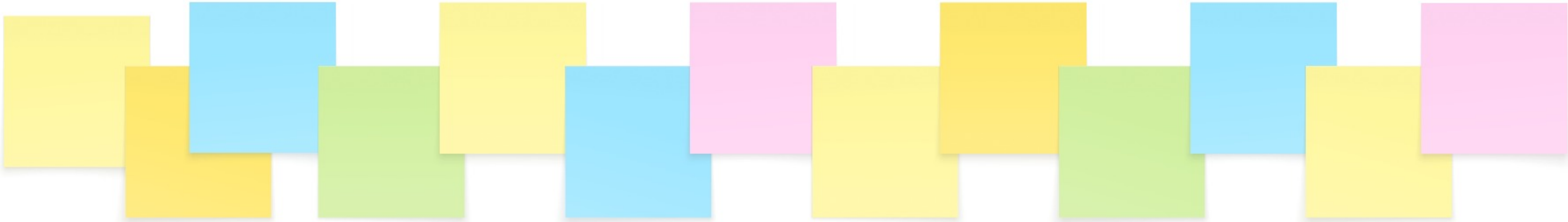
Entity description	Movie Director		
Database	Movie		
Entity name	Movie Director		
Key	Attribute	Data Type	Constraints/comments
FK	Diirector	VARCHAR (50)	Not NULL, Unique
FK	Movie ID	INT	Not Null



Task 2

Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity 1			
Entity description	Director		
Database	Movies		
Entity name	Director		
Key	Attribute	Data Type	Constraints/comments
Primary Key	Director ID	Int	Not Null
	Director Name	VARCHAR(50)	Not Null





Task 1

Identify and name potential entities from the information supplied.

There are at least 5 entities involved with 2 of them 2 are thinking partners

- 1. Customer
- 2. Author
- 3. Editor
- 4. Search(Author, Editor)
- 5. Document, 2 are thought

Empty description

Empty description	Channel ID	Status	Empty value
	Channel ID	Status	Empty value
	Channel ID	Status	Empty value

Navigation bar with a red button and a search bar.

Sticky notes at the bottom of the screen.

Table 1

Entity description	Entity address	Entity name	Entity type
Entity 1	12345	Entity 1	Entity
Entity 2	67890	Entity 2	Entity
Entity 3	54321	Entity 3	Entity

Tablet screen content:

Name	Age	Sex	Transsexual status
John	25	Male	Not transsexual
Jane	30	Female	Not transsexual
John	28	Male	Not transsexual
Jane	22	Female	Not transsexual
John	35	Male	Not transsexual
Jane	27	Female	Not transsexual
John	32	Male	Not transsexual
Jane	29	Female	Not transsexual
John	31	Male	Not transsexual
Jane	26	Female	Not transsexual
John	33	Male	Not transsexual
Jane	24	Female	Not transsexual
John	34	Male	Not transsexual
Jane	23	Female	Not transsexual
John	36	Male	Not transsexual
Jane	21	Female	Not transsexual

Sticky notes below the tablet:

- Yellow sticky note: 1
- Blue sticky note: 2
- Green sticky note: 3
- Yellow sticky note: 4
- Blue sticky note: 5
- Pink sticky note: 6
- Yellow sticky note: 7
- Green sticky note: 8
- Blue sticky note: 9
- Pink sticky note: 10

[illegible]

Table 1

Category	Item	Status	Action
Category 1	Item 1.1	Active	Review
Category 2	Item 2.1	Inactive	Archive
Category 3	Item 3.1	Pending	Approve

Yellow Blue Green Yellow Blue Pink Yellow Green Blue Pink

Step 11

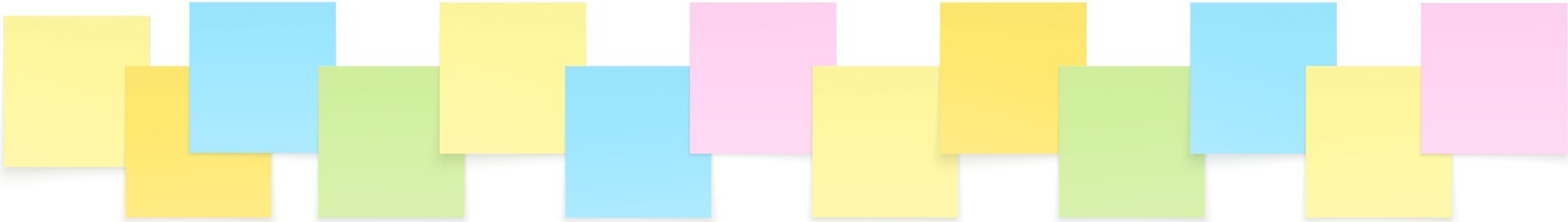
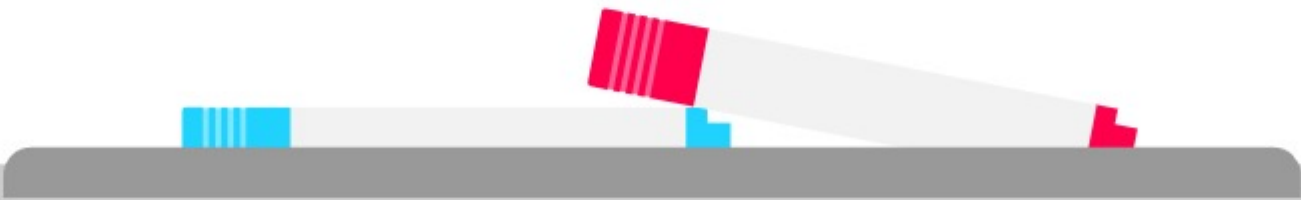
Identify entity relationships. Draw an **entity-relationship diagram** showing the relationships, primarily between the entities, of your database table.

```
graph TD; Living[Living] -.-|Lives in| NYC[New York City]; Wife[Wife] --|is| LA[Los Angeles]; NYC --|is| CH[Chicago]; CH --|is| LA;
```

Entity 2

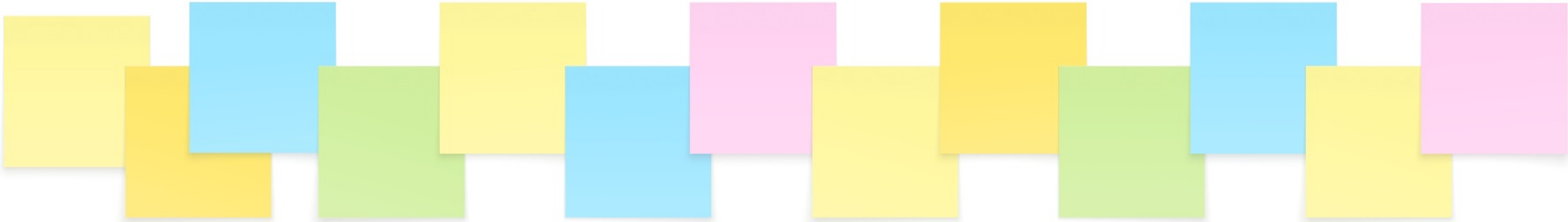
Entity 1

Entity description	Actor Details		
Database	Movies		
Entity name	Actor		
Key	Attribute	Data Type	Constraints/comments
PK	ActorID	Int	Not Null
	ActorName	Varchar(50)	Not Null
	Gender	Char	Not Null M/F/N



Entity 3

Entity description	Film details		
Database	Movies		
Entity name	Movie		
Key	Attribute	Data Type	Constraints/comments
PK	Movie_id	Int	not null
	title	varchar (255)	not null
	budget	money	not null
	release_date	date	not null
	runtime	int	not null
	revenue	money	not null
	movie_status	nvarchar(50)	not null
	vote_average	decimal	null
	vote_count	int	null
	homepage	nvarchar(200)	null
	overview	nvarchar(1000)	null
	tagline	nvarchar(100)	null
	popularity	float	null

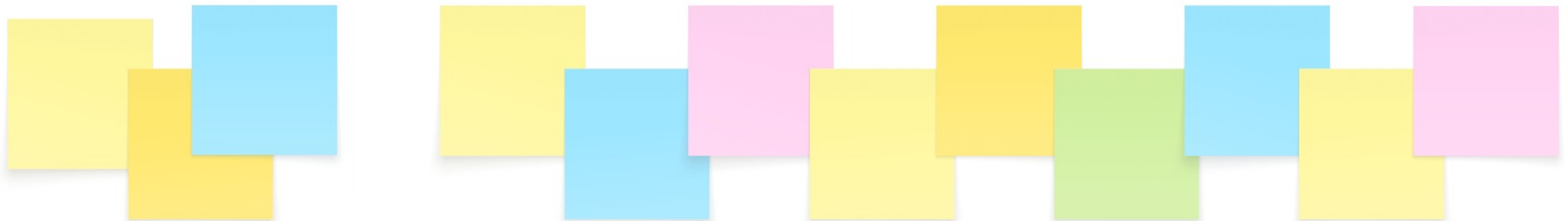


Task 1

Identify and name potential entities from the information supplied.

There are at least 5 entities needed and 2 of these are bridging tables.

1. Director
2. Actor
3. Movie
4. Cast (Actor - Movie)
5. Director - Film Bridge



Movies and Actors Database

Director	Actor	gender	character_name	cast_order	movie_id	title	budget	homepage	overview	popularity	release_date	revenue	runtime	movie_status	tagline	vote_average	vote_count
Quentin Tarantino	Tim Roth	Male	Ted the Bellhop	43	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Tim Roth	Male	Ted the Bellhop	44	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Tim Roth	Male	Ted the Bellhop	45	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Alexandre Rockwell	Tim Roth	Male	Ted the Bellhop	46	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Quentin Tarantino	Antonio Banderas	Male	Man	48	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Antonio Banderas	Male	Man	49	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Antonio Banderas	Male	Man	50	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Alexandre Rockwell	Antonio Banderas	Male	Man	51	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Quentin Tarantino	Jennifer Beals	Female	Angela	52	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Jennifer Beals	Female	Angela	53	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Jennifer Beals	Female	Angela	54	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65

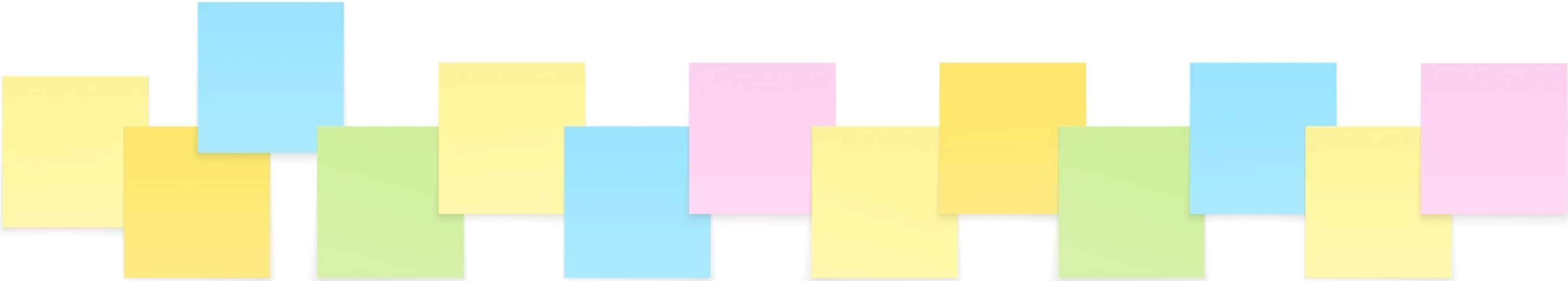
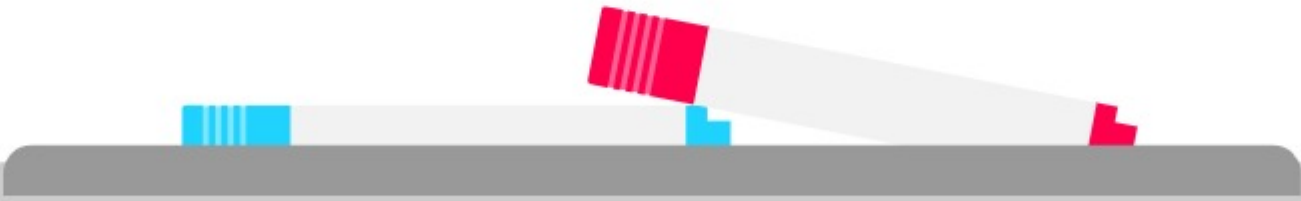
The attributes in the Movies and Actors table are:

- Director
- Actor
- gender
- character_name
- cast_order
- movie_id
- title
- budget
- homepage
- overview
- popularity
- release_date
- revenue
- runtime
- movie_status
- tagline
- vote_average
- vote_count



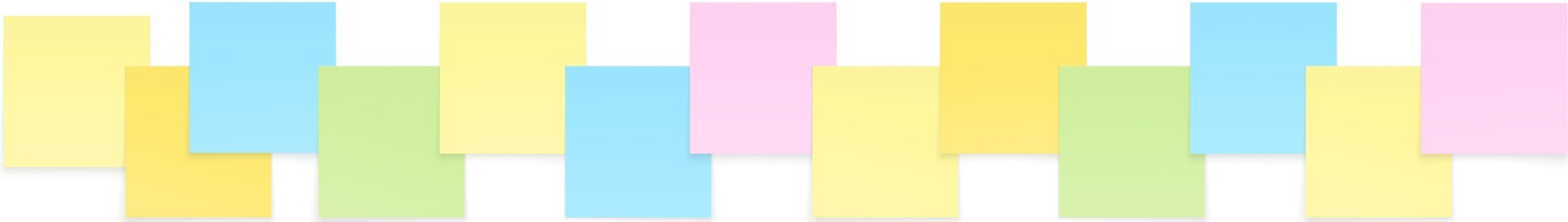
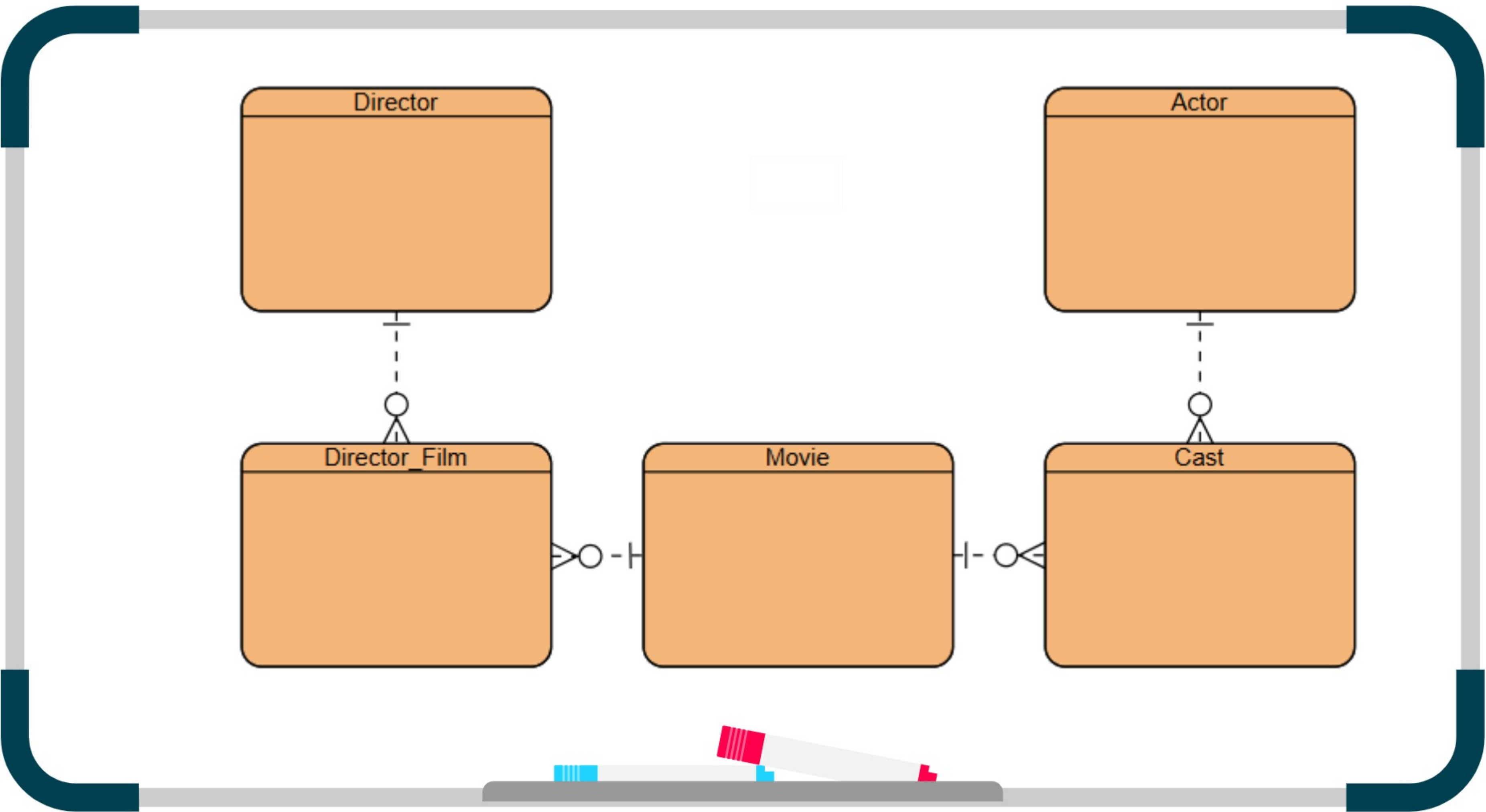
Entity 1

Entity description	Bridging		
Database	Movies		
Entity name	Director_Film		
Key	Attribute	Data Type	Constraints/comments
PK/FK	Director_id	INT	Not null
PK/FK	Movie_id	INT	Not null



Task 3

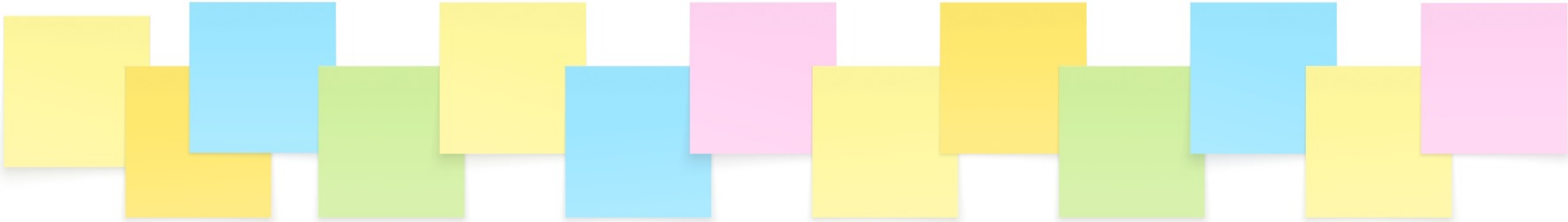
Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities (tables) you identified in Task 1.



Entity 4

Entity 1

Entity description	Cast details		
Database	Movies		
Entity name	Cast		
Key	Attribute	Data Type	Constraints/comments
FK	MovieId	int	not null
FK	ActorId	int	not null
	CharacterName	varchar (50)	not null
	CastOrder	int	not null

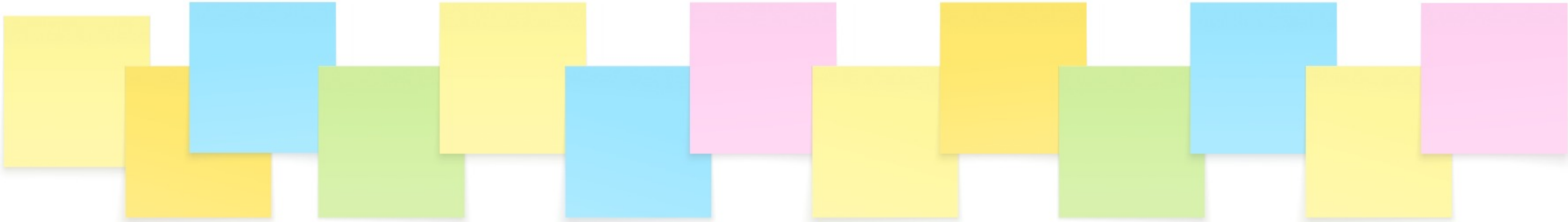


Task 2

Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity 1

Entity description	Director List		
Database	Movies		
Entity name	Director		
Key	Attribute	Data Type	Constraints/comments
PK	Director Id	int	not null
	Director Name	varchar (50)	not null



Task 1

Identify the components of the system and their functions.

1. **Divisor**
 2. **Router**
 3. **Router**
 4. **Router/Multi-Router (Bridge)**
 5. **Router (Bridge)**

Task 2

Identify the components of the system and their functions.

1. **Divisor**
 2. **Router**
 3. **Router**
 4. **Router/Multi-Router (Bridge)**
 5. **Router (Bridge)**

Task 3

Identify the components of the system and their functions.

1. **Divisor**
 2. **Router**
 3. **Router**
 4. **Router/Multi-Router (Bridge)**
 5. **Router (Bridge)**

Task 4

Identify the components of the system and their functions.

1. **Divisor**
 2. **Router**
 3. **Router**
 4. **Router/Multi-Router (Bridge)**
 5. **Router (Bridge)**

Task 5

Identify the components of the system and their functions.

1. **Divisor**
 2. **Router**
 3. **Router**
 4. **Router/Multi-Router (Bridge)**
 5. **Router (Bridge)**

Task 6

Identify the components of the system and their functions.

1. **Divisor**
 2. **Router**
 3. **Router**
 4. **Router/Multi-Router (Bridge)**
 5. **Router (Bridge)**

Task 7

Identify the components of the system and their functions.

1. **Divisor**
 2. **Router**
 3. **Router**
 4. **Router/Multi-Router (Bridge)**
 5. **Router (Bridge)**

Task 8

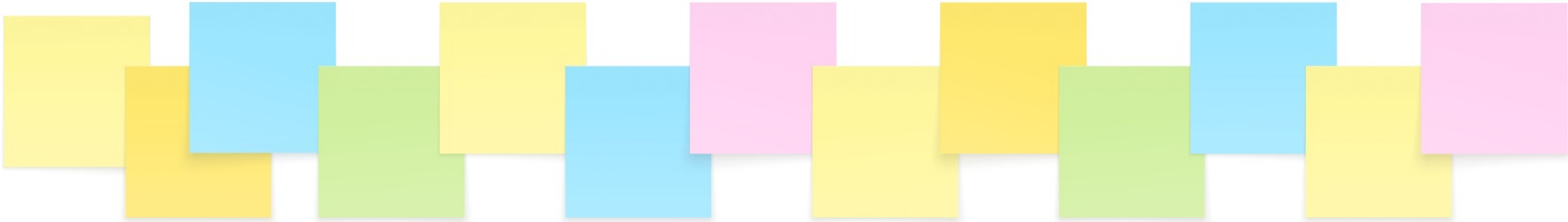
Identify the components of the system and their functions.

1. **Divisor**
 2. **Router**
 3. **Router**
 4. **Router/Multi-Router (Bridge)**
 5. **Router (Bridge)**

Entity 2

Entity 1

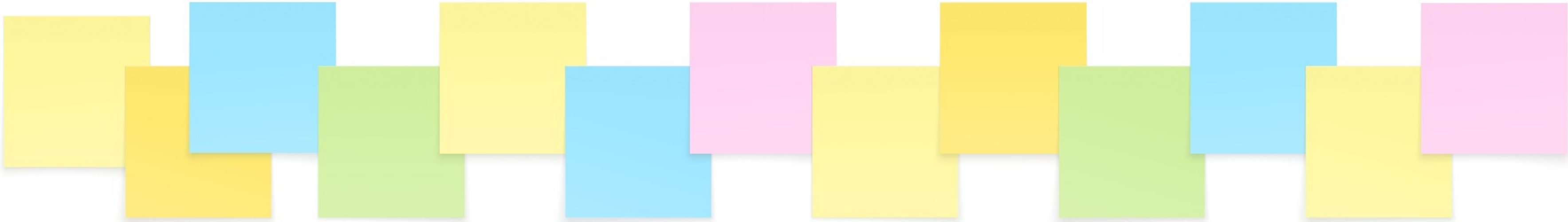
Entity description	List of Actors		
Database	Movies_2		
Entity name	Actor		
Key	Attribute	Data Type	Constraints/comments
Primary Key	Actor_ID	Int	Not Null
	Actor	VarChar (50)	Not Null
	Gender	VarChar (6)	Null



Entity 3

Entity 1

Entity description	List of movies each director is connected to. (Bridge)		
Database	Movies_2		
Entity name	Movie_Director		
Key	Attribute	Data Type	Constraints/comments
Foreign Key	Movie_ID	INT	not null
Foreign Key	Director_ID	INT	not null

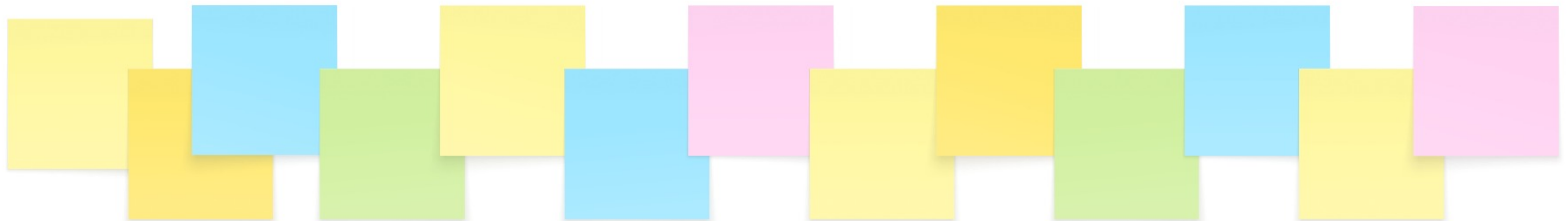


Task 1

Identify and name potential entities from the information supplied.

There are at least 5 entities needed and 2 of these are bridging tables.

1. Director
2. Actor
3. Movie
4. Director/Movie (Bridge)
5. Character (Bridge)

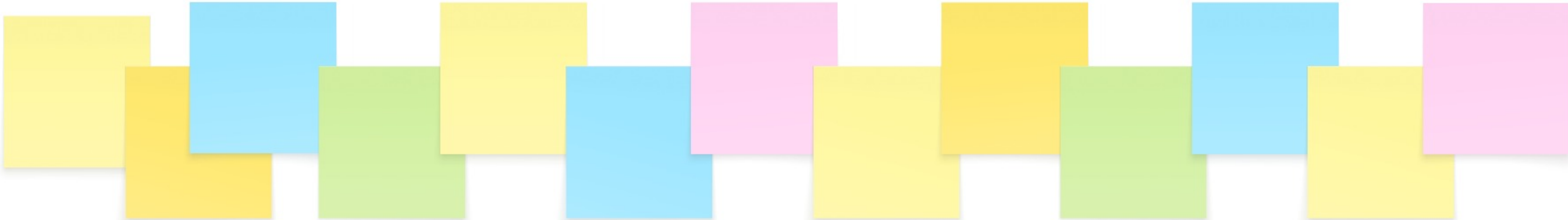


Movies and Actors Database

Director	Actor	gender	character_name	cast_order	movie_id	title	budget	homepage	overview	popularity	release_date	revenue	runtime	movie_status	tagline	vote_average	vote_count
Quentin Tarantino	Tim Roth	Male	Ted the Bellhop	43	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Tim Roth	Male	Ted the Bellhop	44	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Tim Roth	Male	Ted the Bellhop	45	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Alexandre Rockwell	Tim Roth	Male	Ted the Bellhop	46	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Quentin Tarantino	Antonio Banderas	Male	Man	48	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Antonio Banderas	Male	Man	49	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Antonio Banderas	Male	Man	50	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Alexandre Rockwell	Antonio Banderas	Male	Man	51	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Quentin Tarantino	Jennifer Beals	Female	Angela	52	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Robert Rodriguez	Jennifer Beals	Female	Angela	53	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65
Allison Anders	Jennifer Beals	Female	Angela	54	11	Star Wars	11000000	http://www.starwars.com/films/star-wars-episode-iv-a-new-hope	In a futuristic city sharply divided between the	32.351527	10/01/1927	650422	153	Released	There can be no understanding between the hand	8	65

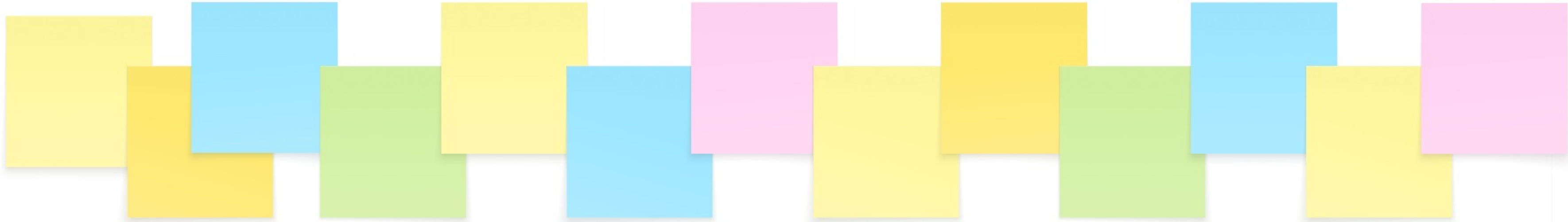
The attributes in the Movies and Actors table are:

- Director
- Actor
- gender
- character_name
- cast_order
- movie_id
- title
- budget
- homepage
- overview
- popularity
- release_date
- revenue
- runtime
- movie_status
- tagline
- vote_average
- vote_count

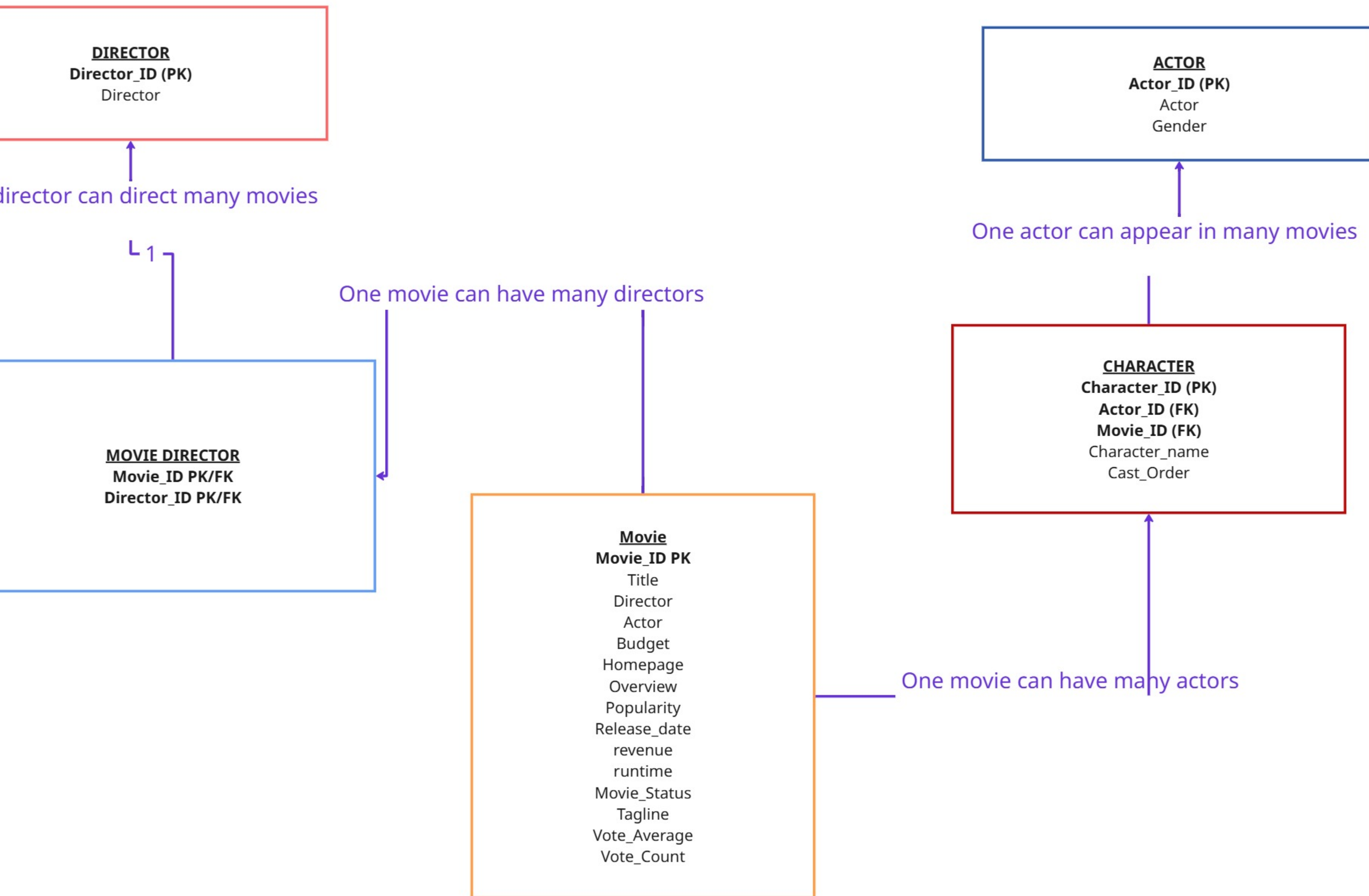


Entity 5

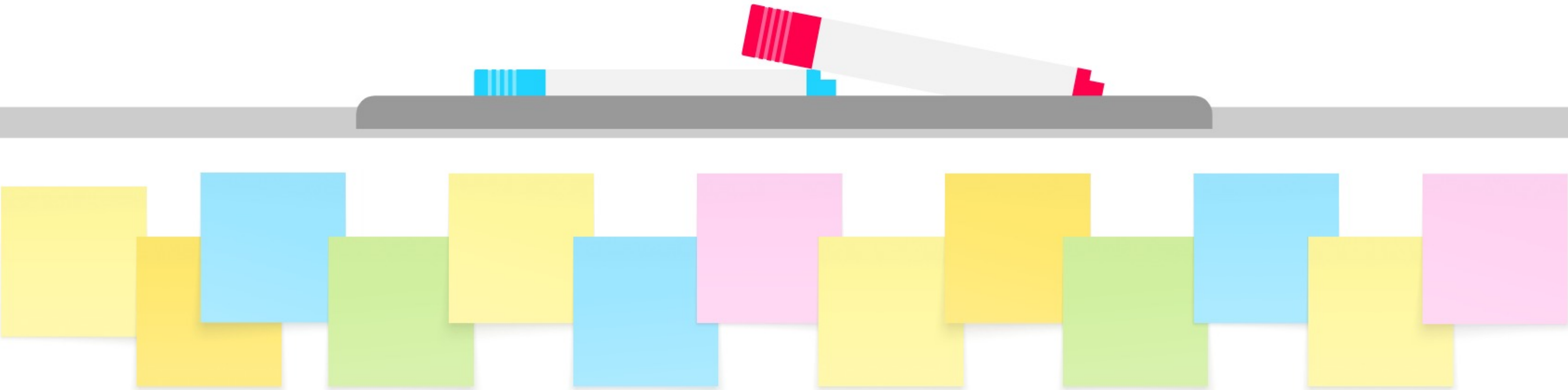
Entity 1			
Entity description	How characters link to Movies and Actors. (bridge)		
Database	Movies_2		
Entity name	Character		
Key	Attribute	Data Type	Constraints/comments
Foreign Key	Movie_ID	INT	not null
Foreign Key	Actor_ID	INT	not null
Primary Key	Character_ID	INT	not null
	Character_Name	VarChar(50)	not null
	Cast_Order	INT	not null



relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities identified in Task 1.



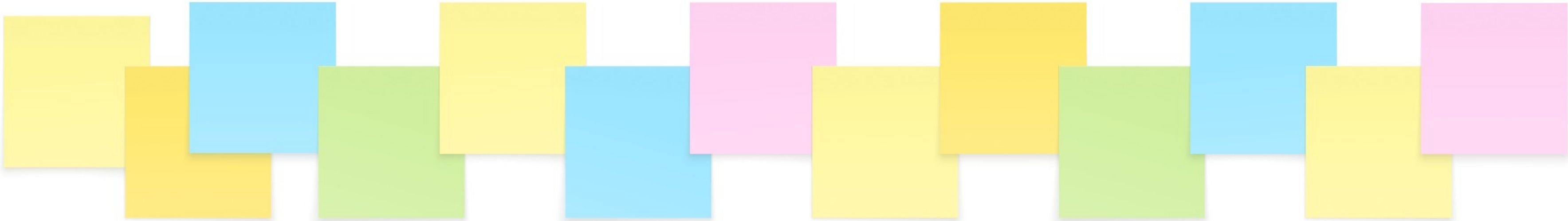
Entity description	List of Movies		
Database	Movies_2		
Entity name	Movie		
Key	Attribute	Data Type	Constraints/comments
Primary Key	Movie_ID	INT	not null
	Title	nvarChar(max)	not null
Foreign Key	Director_ID	INT	not null
Foreign Key	Actor_ID	INT	not null
	Budget	Money	null
	Homepage	nvarchar(max)	null
	Overview	nvarchar(max)	null
	Popularity	Float	null
	Release_Date	date	null
	Revenue	INT	null
	Runtime	INT	null
	Movie_Status	VarChar(50)	not null
	Tagline	nvarchar(max)	null
	Vote_Average	decimal (3,2)	null
	Vote_Count	INT	null



Task 2

Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity 1			
Entity description	List of Director		
Database	Movies_2		
Entity name	Director		
Key	Attribute	Data Type	Constraints/comments
Primary Key	Director_ID	Int	NOT NULL
	Director	Vchar (50)	NOT NULL



Activity 1: Movies Database Normalisation

Learner instructions

We have been given the task of creating an online resource based on movies produced in the last 50 years. One of the features of this resource will be a RDMS that will store details of the personnel and production information relating to a large number of films. We have an Excel file provided with data that has been partly normalised in that some of the entities and their attributes have already been identified. However one of the sheets on the Excel file labelled Movies and Actors still needs to be normalised before it is suitable for the RDMS. Screenshots can be seen below of this table and its attributes.

Task 1

The attributes in the Movies and Actors table are:

- Director
- Actor
- gender
- character_name
- cast_order
- movie_id
- title
- budget
- homepage
- overview
- popularity
- release_date
- revenue
- runtime
- movie_status
- tagline
- vote_average
- vote_count

Identify and name potential entities from the information supplied. Remember that there may be many-to-many relationships between the tables that would require a bridging table (a table containing the foreign keys of the tables to be joined) Think about the fact that and actor can play many characters but these characters can appear in a number of movies i.e. Luke Skywalker appears in Star Wars but also in The Empire Strikes Back! Also when you create any of the main tables from they will need a Primary Key attribute that may need to be added for example director_ID Identify and name potential entities from the information supplied. Hint: There are at least 5 entities needed and 2 of these are bridging tables.

1. **Director**
2.
3.
4.
5.

Task 2

Compile data dictionaries for your identified entities. You may use the pro forma on the following page (or any other valid format). You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity description			
Database			
Entity name			
Key	Attribute	Data Type	Constraints/comments

Task 3

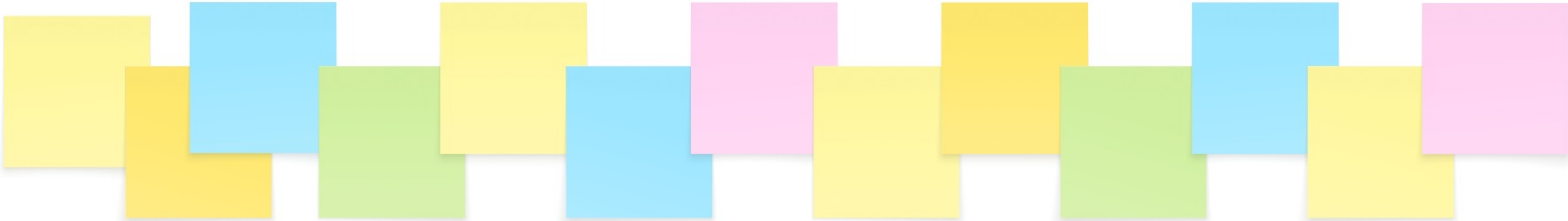
Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities(tables) you identified in Task 1. Show the Primary and Foreign keys to show how tables would relate to each other i.e. The Primary key of one table (Parent) would become the Foreign key in its related table (Child) Hint: Bridging tables that are used to avoid many-to-many relationships are basically formed of two one-to-many relationships. The diagram below shows the “many-to-many” relationship to avoid duplicate values or multiple values in the one column, between Orders and Products. OrderID and ProductID have become foreign keys in the OrderDetails table.

Replace the image above with your own Entity Relationship Diagram (ERD)
[Entity Relationship Diagram: Untitled | Visual Paradigm Online](#)

Entity 5

Entity 1

Entity description	Links Movie and Actors, storing cast details		
Database	Movie		
Entity name	MovieCast		
Key	Attribute	Data Type	Constraints/comments
Primary Key/Foreign Key	Movie_ID	Int	Not null
Primary Key/Foreign Key	Actor_ID	Int	Not null
	Character_Name	Varchar(100)	Null Allowed
	Cast_Order	Int	Null Allowed



Entity 4

Entity 1

Entity description	Links Movie and Directors		
Database	Movie		
Entity name	MovieDirector		
Key	Attribute	Data Type	Constraints/comments
Primary Key/Foreign Key	Movie_ID	Int	Not null
Primary Key/Foreign Key	Director_ID	Int	Not null

